# Problem J. Find the Gap

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|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 second |
| Memory limit: | 512 megabytes |

You are given $n$ points in the 3D space. Please find two parallel planes such that all the $n$ points are inside the gap of the two parallel planes, and the length of the gap is minimized.

## Input

The first line of the input contains a single integer $n$ ($1 \le n \le 50$), denoting the number of points.

Each of the following $n$ lines contains three integers $x_i$, $y_i$ and $z_i$ ($1 \le x_i, y_i, z_i \le 10\,000$), describing a point $(x_i, y_i, z_i)$. It is guaranteed that all the $n$ points are pairwise distinct.

## Output

Print a single line containing a single real number: the minimum possible length of the gap with an absolute or relative error of at most $10^{-9}$.

Precisely speaking, assume that your answer is $a$ and the jury's answer is $b$. Your answer will be considered correct if and only if $\frac{|a-b|}{\max\{1,|b|\}} \le 10^{-9}$.

## Examples

| standard input | standard output |
|---|---|
| 8<br>1 1 1<br>1 1 2<br>1 2 1<br>1 2 2<br>2 1 1<br>2 1 2<br>2 2 1<br>2 2 2 | 1.000000000000000 |
| 5<br>1 1 1<br>1 2 1<br>1 1 2<br>1 2 2<br>2 1 1 | 0.707106781186548 |